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Adherence to the Mediterranean diet and its association with atherogenic indices and lipid profile in individuals seeking a weight-loss dietary program: what role for body fat?

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Abstract

Background: The Mediterranean diet is widely recognized for its cardiovascular benefits, but its specific effects on atherogenic indices remain unclear—particularly in individuals seeking a weight-loss dietary program, where excess fat mass may mitigate the diet's protective effects.

Methods: To explore this relationship, a cross-sectional study was conducted involving 10,286 participants enrolled in a weight-loss dietary program. Anthropometric and biochemical data were collected, and adherence to the Mediterranean diet was assessed using the 14-item Mediterranean Diet Adherence Screener questionnaire. Lipid profiles—including total cholesterol, LDL, HDL, and triglycerides—were analyzed to calculate the following atherogenic indices: atherogenic index of plasma, Castelli risk indices I and II, lipoprotein combine index, atherogenic coefficient, and atherogenic combined index.

Results: Multivariate linear regression models—adjusted for sex, age, body mass index, smoking, physical activity, sociodemographic factors, and use of lipid-lowering medications—showed that each one-point increase in the Mediterranean diet adherence score was significantly associated with reductions in the atherogenic index of plasma (-0.003 , 95%CI: 0.006 , -0.000), the atherogenic coefficient (-0.013 , 95%CI: -0.024 , -0.001), Castelli risk index I (-0.013 , 95%CI: -0.024 , -0.001), and the atherogenic combined index (-0.004 , 95%CI: -0.007 , -0.000), with a marginal association observed for Castelli risk index II (-0.009 , 95%CI: -0.019 , 0.000). Significant associations were also observed for total cholesterol (-3.453 mg/dl, 95%CI: -5.911 , -0.995), LDL (-3.225 mg/dl, 95%CI: -5.402 , -1.048), and HDL concentrations ($+0.255$ mg/dl, 95%CI: 0.098 , 0.412). However, except for HDL, these associations lost statistical significance after adjusting for body fat percentage. Significant interactions between Mediterranean diet adherence score and body fat percentage were observed for several atherogenic indices, including the

atherogenic coefficient, Castelli risk index I, Castelli risk index II, and the atherogenic combined index as well as for total cholesterol and LDL, suggesting that the protective effects of the Mediterranean diet diminish as fat mass increases.

Conclusions: Although adherence to the Mediterranean diet is associated with more favorable lipid profiles and atherogenic indices, these benefits are modulated by body composition, particularly fat mass. These findings highlight the importance of integrated dietary strategies that combine nutritional quality with body fat reduction to support cardiovascular prevention.

Keywords: Mediterranean Diet; Dyslipidemias; Body Fat; Obesity

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List of abbreviations

AI – Atherogenic Indices

AIP – Atherogenic Index of Plasma

CRI-I – Castelli Risk Index I

CRI-II – Castelli Risk Index II

LCI – Lipoprotein Combine Index

AC – Atherogenic Coefficient

ACI – Atherogenic Combined Index

TG – Triglycerides

TC – Total cholesterol

CVD – Cardiovascular Disease

VAT – Visceral Adipose Tissue

Introduction

In 2021, the global number of cardiovascular disease (CVD) cases had reached 523 million, nearly doubling from 271 million in 1990 [1]. This gradual yet substantial increase has made CVD the leading cause of morbidity, disability, and mortality worldwide [2]. According to the European Society of Cardiology, CVDs account for approximately 45% of all deaths in Europe and 34% of total mortality in Italy [3]. In addition to the serious human consequences, there is a significant economic burden: current estimates place the indirect annual costs of \$237 billion in 2015, with projections rising to \$368 billion by 2035 [4].

CVDs are multifactorial conditions resulting from a complex interplay of genetic, lifestyle, and environmental factors [1]. A key pathophysiological mechanism underlying CVD is atherosclerosis, a progressive process characterized by the buildup of plaques in arterial walls that can begin as early as childhood. Although initially asymptomatic, atherosclerotic plaques may eventually grow large enough to impair blood flow, leading to major clinical outcomes such as coronary artery disease, cerebrovascular disease, peripheral artery disease, and aortic atherosclerosis which collectively represent the four major manifestations of CVD [5].

Traditionally, the assessment of atherosclerotic risk has relied on clinical history and standard lipid parameters such as total cholesterol, LDL-C, HDL-C, and triglycerides. Although these markers remain fundamental for cardiovascular risk stratification, they may not fully capture the complexity of lipid-related atherogenesis [6]. In recent years, several composite lipid parameters, collectively known as atherogenic indices (AI), have been proposed to provide a more integrated view of lipid-related risk [7]. These indices, including the Atherogenic Index of Plasma (AIP), Castelli's Risk Indices (CRI-I and CRI-II), the Atherogenic Coefficient (AC), the Lipoprotein Combine Index (LCI), and Atherogenic Combined Index (ACI) combine multiple lipid variables into single measures. Emerging

evidence suggests that atherogenic indices may complement traditional lipid parameters and provide additional information on cardiovascular risk [8-10]. For example, one study reported that AIP, LCI, AC, CRI-I, and CRI-II were stronger predictors of the coronary artery aneurysm risk than conventional lipid parameters [8]. Another study found that AIP better predicted mortality risk than individual cholesterol measures in older adults [10]. Additionally, AIP has been shown to predict long-term cardiovascular outcomes in community-based cohorts independently of traditional lipid parameters [9].

Numerous modifiable factors contribute to the development of atherosclerosis, among which diet plays a central role [11]. In particular, the Mediterranean diet, characterized by high consumption of fruits, vegetables, legumes, whole grains, fish, low-fat dairy products, nuts, seeds, and extra virgin olive oil as the main source of fat, has been widely recognized for its cardioprotective properties [12, 13]. Several meta-analyses have demonstrated that adherence to the Mediterranean diet is associated with reduced CVD risk and improvements in cardiovascular risk factors [12-14], including reductions in body fat [14], total cholesterol levels [15], vascular function [16], systemic inflammation and the progression of subclinical atherosclerosis [17]. However, its specific impact on lipid-derived atherogenic indices remains unclear, as no studies to date have directly examined this association. This question is particularly relevant in individuals at elevated cardiovascular risk, such as those following a weight-loss dietary intervention. In these individuals, excess adiposity contributes to chronic inflammation, insulin resistance, and disturbances in lipid metabolism through multiple interrelated mechanisms, including increased release of free fatty acids, elevated proinflammatory cytokine production, and enhanced hepatic VLDL-C synthesis, which may collectively attenuate the protective effects of the Mediterranean diet [18, 19]. It was therefore hypothesized that greater adherence to the Mediterranean diet would be associated with more favorable atherogenic indices and lipid profiles in

individuals seeking a weight-loss dietary program. Furthermore, body fat was hypothesized to play a key role in this relationship by acting either as a mediator or an effect modifier, potentially attenuating the cardioprotective effects of the Mediterranean diet in individuals with higher adiposity.

Therefore, the aim of this study was to evaluate the relationship between adherence to the Mediterranean diet and AI and lipid profiles in a large cohort of individuals seeking a weight-loss dietary program, while accounting for body fat, thereby providing novel evidence on the role of adiposity in modulating diet–atherogenic risk relationships.

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Material and methods

Study design and sample selection

A total of 10,286 participants were recruited from those attending the International Center for the Assessment of Nutritional Status (ICANS) at the University of Milan from September 2010 to September 2022. Enrollment was restricted to individuals aged over 18 who did not meet any of the exclusion criteria: diabetes, cardiovascular disease, cancer within the past 5 years, hepatitis C virus infection, hepatitis B virus infection, liver cirrhosis, neurological disease, gastrointestinal disease, cardiac failure, renal failure, pulmonary insufficiency and use of drugs known to cause lipodystrophy. At recruitment, all participants underwent a series of assessments, starting with a medical interview to evaluate their medical history, current drug therapies, smoking habits, and physical activity. Physical activity was assessed by asking participants whether they practiced any structured activity and, if so, how many hours per week they spent on it. Individuals reporting ≥ 2 hours per week of structured physical activity were classified as active [20]. Additionally, participants underwent a nutritional evaluation, during which anthropometric measurements were taken, and a fasting blood sample was collected. Finally, all participants were instructed to complete a validated questionnaire to assess their adherence to the Mediterranean diet. The study was conducted in accordance with the guidelines of the Helsinki Declaration, and each participant read and signed a written informed consent. The Ethics Committee of the University of Milan approved the study procedures (study protocol No. 23/2016).

Anthropometric measurements

All anthropometric data were collected based on the international guidelines. Before undergoing all measurements, participants were asked to wear only underwear and socks.

Weight was measured using a column scale with a precision of 0.1 kg while height was assessed using a stadiometer and rounded to the nearest 0.5 cm. Body Mass Index (BMI) was then calculated by dividing weight (kg) by the square of height (m²) and categorized following World Health Organization (WHO) standards: underweight (BMI <18.5 kg/m²), normal weight (BMI 18.5–25 kg/m²), overweight (BMI 25–29.9 kg/m²), or obese (BMI ≥30 kg/m²). Waist circumference was measured with a non-elastic tape placed horizontally between the lower border of the rib cage and the iliac crest. Additionally, skinfold thickness at the triceps, biceps, subscapular, and suprailiac sites was recorded using a plicometer with a precision of 0.2 mm. For each site, the measurements were averaged from three consecutive readings. Following international standards, skinfolds were measured on the non-dominant side of each participant. Body density (ρ) was estimated using the equations from Durnin and Womersley [21], which account for both gender and age. Body fat percentage was calculated using Siri's formula: $\text{Body fat \%} = (4.95 / \rho - 4.50) \times 100$.

Adherence to the Mediterranean diet

The 14-item MEDAS questionnaire was employed to evaluate adherence to the Mediterranean Diet, a short dietary screening tool validated in the PREMED trial [22]. This questionnaire consists of 14 questions designed to assess the frequency of consumption of foods typical of both Mediterranean and Western diets. A point is given for each food item that aligns with the Mediterranean diet criterion. Briefly, 1-point was attributed for each of the following: (1) olive oil as the main cooking fat; (2) olive oil ≥ 4 tablespoons/day; (3) vegetables ≥ 2 servings/day; (4) fruit ≥ 3 servings/day; (5) red or processed meat < 1 serving/day; (6) butter or cream or margarine < 1/day; (7) sugar-sweetened beverages < 1/day; (8) wine ≥ 3 glasses/week; (9) legumes ≥ 3 servings/week; (10) fish/seafood ≥ 3 servings/week; (11) commercial sweets and confectionery < 3/week;

(12) nuts ≥ 1 /week; (13) white more than red meats (yes) and; (14) use of soffritto ≥ 2 /week [23, 24]. The total score, ranging from 0 to 14, is the sum of the individual points.

Blood sampling, lipid profile and atherogenic indices

A fasting blood sample was collected between 8:30 and 9:30 a.m. Fasting total cholesterol (TC), HDL-cholesterol (HDL-C), LDL-cholesterol (LDL-C) and triglycerides (TG) were measured using enzymatic methods (Cobas Integra 400 Plus; Roche Diagnostics). The resulting values were used to calculate atherogenic indices (AI) using the following formulas:

$$AIP = \log_{10}\left(\frac{TG}{HDL - C}\right)$$

$$LCI = \frac{TC \times TG \times LDL - C}{HDL - C}$$

$$CRI - I = \frac{TC}{HDL - C}$$

$$CRI - II = \frac{LDL - C}{HDL - C}$$

$$AC = \frac{TC - HDL - C}{HDL - C}$$

$$ACI = \log_{10}\left(\frac{TG \times non - HDL - C}{HDL - C}\right)$$

For AIP and LCI, lipid parameters were entered in mmol/L, whereas for the other indices they were expressed in mg/dL.

Statistical Analysis

Continuous variables are presented as 50th (median), 25th and 75th (interquartile range, IQR) percentiles due to the non-normal distribution of some variables, as verified by the Kolmogorov–Smirnov test. Categorical variables are reported as frequencies and

percentages. The association between the MEDAS score and the outcomes of interest (atherogenic indices and lipid profile parameters) was assessed using a pre-specified sequence of linear regression models defined a priori based on biological plausibility:

Model 1 (M1): Unadjusted

Model 2 (M2): Adjusted for sex (binary: 0 = female, 1 = male) and age (continuous, in years)

Model 3 (M3): M2 plus adjustments for BMI (continuous, kg/m²), smoking status (categorical: 0 = never smoker, 1 = former smoker, 2 = current smoker), physical activity (binary: 0 = no, 1 = ≥ 2 hours/week), education level (categorical: 0 = primary/lower secondary, 1 = high school diploma, 2 = university degree or higher), marital status (categorical: 0 = single, 1 = married, 2 = divorced/widowed), and use of lipid-lowering therapy (binary: 0 = no, 1 = yes)

Model 4 (M4): M3 plus body fat percentage (continuous)

Model 5 (M5): Given the presence of significant interactions between MEDAS score and body fat percentage for several outcomes, the interaction term (MEDAS score#body fat) was included in the model when statistically significant

Model 1 tests the unadjusted association between adherence to the Mediterranean diet and the outcomes of interest. Model 2 evaluates this association accounting for sex and age. Model 3 further adjusts for BMI, sociodemographic factors, and lipid-lowering treatment. Model 4 incorporates body composition into the analysis. This model investigates whether fat mass acts as a mediator of the association between adherence to the Mediterranean diet and the outcomes of interest. Finally, Model 5 assesses the effect of adherence to the Mediterranean diet in relation to body composition, while controlling for

all other potential confounders. This model investigates whether fat mass acts instead as a modifier of the association between adherence to the Mediterranean diet and the outcomes of interest. Collinearity was assessed using the Variance Inflation Factor (VIF) for each predictor and for the overall model; all VIF values were <5 , indicating negligible multicollinearity. Homoscedasticity and potential violations of the linearity assumption between continuous predictors and the dependent variable were assessed both graphically, using residual-versus-fitted plots, and through statistical tests. The Breusch-Pagan test was applied to check for heteroscedasticity, and robust confidence intervals were calculated for all models where heteroscedasticity was detected. Multivariable fractional polynomials were used to model potential non-linear associations between continuous predictors and the outcome. Several transformations of age and BMI were selected depending on the model. However, there was only a modest gain in the linearity of continuous predictors when transformations were applied; therefore, to preserve interpretability and readability, all continuous covariates were kept in their original, untransformed form. The significance of interactions was formally tested using Wald tests, and all interactions reported in the manuscript are statistically supported. To address potential overadjustment, additional analyses were performed using a more parsimonious model that included only variables remaining significant (Supplementary tables S1-S2). Results were consistent with those of the main analyses, suggesting that overadjustment was unlikely. In addition, we evaluated the interaction between MEDAS and sex. Analyses showed that the MEDAS#body fat and MEDAS#sex interactions were both significant when tested separately, with identical R^2 values, suggesting that they explained the same proportion of variability. Moreover, when the two interaction terms were included simultaneously in the same model, both lost statistical significances, indicating redundancy between the modifiers. In light of this, and considering that body fat showed a greater number of significant interactions and is a continuous and modifiable variable, we decided

to report the association between MEDAS, atherogenic indices, and lipid parameters in relation to body fat. A *P* value < 0.05 was considered statistically significant. Statistical analyses were performed using STATA software, version 18.5 (StataCorp LP).

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Results

The study included 10,286 participants, of whom 72.7% were women and 27.3% were men. The median age and BMI were 46 years (IQR: 37–55) and 28 kg/m² (IQR: 25.1–31.6), respectively. The prevalence of overweight and obesity was 41.1% and 34.6%, respectively. Table 1 summarizes the characteristics of the study population.

Table 1: Characteristics of patients

	Women n=7477			Men n=2809			Total n=10286		
	Median	P25	P75	Median	P25	P75	Median	P25	P75
Age (years)	46	37	54	46	38	55	46	37	55
BMI (kg/m ²)	27.4	24.5	31	29.5	26.9	33	28	25.1	31.6
Body fat (%)	38.9	35.4	42	31.4	27.5	35	37.2	32.5	40.7
Waist circumference (cm)	91.1	83	100	105	97.1	113	95	85.7	105
Total cholesterol (mg/dl)	207	182	234	207	181	235	207	182	234
LDL cholesterol (mg/dl)	127	105	152	137	114	161	130	107	155
HDL cholesterol (mg/dl)	64	54	74	48	41	56	59	49	71
Triglycerides (mg/dl)	82	61	113	114	81	163	89	65	127
AIP	-0.3	-0.4	-0.1	0	-0.2	0.2	-0.2	-0.4	0
AC	2.2	1.7	2.9	3.3	2.5	4.2	2.4	1.8	3.3
CRI-I	3.2	2.7	3.9	4.3	3.5	5.2	3.4	2.8	4.3
CRI-II	2	1.6	2.6	2.8	2.2	3.5	2.2	1.7	2.9
LCI	10	5.7	17	19.4	10.8	35	11.8	6.4	21.8
ACI	2.3	2	2.5	2.6	2.3	2.8	2.3	2.1	2.6
MEDAS score	6	5	8	6	5	8	6	5	8
	N	%		N	%		N	%	
Physical activity (≥2h/week)	3582	47.9		1445	51.4		5027	48.9	
Smoking									
Not smoker	4121	55.1		1302	46.4		5423	52.7	
Ex smoker	1362	18.2		747	26.6		2109	20.5	
Smoker	1994	26.7		760	27.1		2754	26.8	
Education									
Elementary or middle school	679	9.1		306	10.9		985	9.6	
High school	3528	47.2		1245	44.3		4773	46.4	
Master degree or higher	3270	43.7		1258	44.8		4528	44	
Marital status									
Not married	2989	40		1042	37.1		4031	39.2	
Married	3772	50.4		1570	55.9		5342	51.9	
Divorced or widower	716	9.6		197	7		913	8.9	
Lipid-lowering drug therapy	238	3.2		186	6.6		424	4.1	

High LDL-C (≥ 116 mg/dl)	4752 63.5	2079 74.0	6831 66.4
High triglycerides (≥ 150 mg/dl)	843 11.3	845 30.1	1688 16.4
Mixed dyslipidemia (High LDL-C + High triglycerides)	674 9.0	684 24.4	1358 13.2

Abbreviations: P25, 25th percentile; P75, 75th percentile

Table 2 presents the main regression models evaluating the association between the MEDAS score and atherogenic indices. Complete regression results for each individual AI are provided in Supplementary Tables S3–S8. After adjustment for sex, age, BMI, lifestyle factors, sociodemographic variables, and lipid-lowering drugs (Model 3), the MEDAS score was significantly associated with lower values of AIP, AC, CRI-I, and ACI, and marginally associated with lower CRI-II values. Specifically, each 1-point increase in the MEDAS score was associated with a reduction of 0.003 points in AIP (95% CI: -0.006 , -0.000 , $P = 0.049$), 0.013 points in AC (95% CI: -0.024 , -0.001 , $P = 0.029$), 0.013 points in CRI-I (95% CI: -0.024 , -0.001 , $P = 0.029$), and 0.004 points in ACI (95% CI: -0.007 , -0.000 , $P = 0.049$). The association with CRI-II was not statistically significant ($\beta = -0.009$; 95% CI: -0.019 , 0.000 ; $P = 0.052$), although a trend consistent with other atherogenic indices was observed. No significant association was found between the MEDAS score and LCI. After further adjustment for body fat percentage (Model 4), all associations were lost statistical significance. However, for four out of six indices, a significant interaction between MEDAS score and body fat percentage (Model 5) was observed. In particular, the strength of the inverse association between the MEDAS score and AI such as AC, CRI-I, CRI-II, and ACI decreased as body fat percentage increased. Figure 1 gives the effects plots corresponding to Model 4 for AIP and LCI, and to Model 5 for AC, CRI-I, CRI-II, and ACI.

Table 2. Association between Mediterranean Diet adherence and atherogenic indices

	AIP	AC	CRI-I	CRI-II	LCI	ACI
	M4	M5	M5	M5	M4	M5
MEDAS score (1-point)	-0.002	-0.091*	-0.091*	-0.083**	-0.02	-0.024*
	[-0.005,0.001]	[-0.160,-0.021]	[-0.160,-0.021]	[-0.140,-0.026]	[-0.223,0.182]	[-0.045,-0.002]
Body fat (%)	0.009***	0.023**	0.023**	0.015*	0.490***	0.010***
	[0.007,0.010]	[0.009,0.037]	[0.009,0.037]	[0.003,0.026]	[0.383,0.596]	[0.006,0.014]
MEDAS score # Body fat		0.002*	0.002*	0.002**		0.001*
		[0.000,0.004]	[0.000,0.004]	[0.001,0.004]		[0.000,0.001]
Sex (Male)	0.299***	1.271***	1.271***	0.978***	16.070***	0.379***
	[0.283,0.316]	[1.202,1.339]	[1.202,1.339]	[0.921,1.035]	[14.755,17.384]	[0.358,0.400]
Age (years)	0	0.005***	0.005***	0.006***	0.114***	0.002***
	[-0.000,0.001]	[0.003,0.007]	[0.003,0.007]	[0.004,0.007]	[0.081,0.146]	[0.002,0.003]
BMI (kg/m ²)	0.010***	0.028***	0.028***	0.022***	0.333***	0.010***
	[0.009,0.012]	[0.022,0.033]	[0.022,0.033]	[0.018,0.027]	[0.238,0.428]	[0.009,0.012]
Physical activity (≥2h/week)	-0.030***	-0.149***	-0.149***	-0.103***	-2.287***	-0.041***
	[-0.039,-0.020]	[-0.187,-0.111]	[-0.187,-0.111]	[-0.135,-0.071]	[-2.958,-1.616]	[-0.053,-0.029]
Smoking (Ex smoker)	0.005	-0.039	-0.039	-0.054**	-0.21	0.002
	[-0.007,0.018]	[-0.088,0.009]	[-0.088,0.009]	[-0.094,-0.013]	[-1.068,0.648]	[-0.013,0.018]
Smoking (Smoker)	0.049***	0.174***	0.174***	0.134***	2.985***	0.056***
	[0.038,0.061]	[0.128,0.219]	[0.128,0.219]	[0.096,0.172]	[2.150,3.819]	[0.042,0.070]
Education (High school)	-0.006	0.031	0.031	0.039	0.042	-0.001
	[-0.022,0.011]	[-0.036,0.099]	[-0.036,0.099]	[-0.017,0.095]	[-1.224,1.308]	[-0.022,0.020]
Education (Master degree)	-0.008	0.004	0.004	0.025	-0.564	-0.01
	[-0.025,0.009]	[-0.064,0.071]	[-0.064,0.071]	[-0.031,0.082]	[-1.839,0.711]	[-0.031,0.011]
Marital status (Married)	-0.007	-0.003	-0.003	-0.002	-0.295	-0.008
	[-0.018,0.004]	[-0.046,0.041]	[-0.046,0.041]	[-0.039,0.034]	[-1.070,0.480]	[-0.022,0.006]
Marital status (Divorced/ widower)	-0.002	-0.021	-0.021	-0.026	-0.298	-0.002
	[-0.020,0.017]	[-0.094,0.052]	[-0.094,0.052]	[-0.088,0.035]	[-1.556,0.960]	[-0.025,0.021]
Lipid-lowering therapy (yes)	0.042**	-0.267***	-0.267***	-0.331***	-2.840**	-0.013
	[0.016,0.068]	[-0.376,-0.157]	[-0.376,-0.157]	[-0.423,-0.239]	[-4.799,-0.881]	[-0.046,0.019]
Constant	-0.852***	0.538*	1.538***	0.702***	-18.131***	1.515***
	[-0.897,-0.808]	[0.051,1.026]	[1.051,2.026]	[0.304,1.100]	[-21.258,-15.004]	[1.364,1.667]

Values are regression coefficients and 95% confidence intervals [in bracket] obtained from multivariate linear regression models.

M4: model adjusted for sex, age, BMI, body fat, smoking status (reference category: never smoker), physical activity (reference category: <2 hours/week), education level (reference category: elementary/middle school), marital status (reference category: single), and use of lipid-lowering therapy (reference category: no)

M5: M4 + MEDAS score#body fat interaction

Abbreviations:

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Figure 1

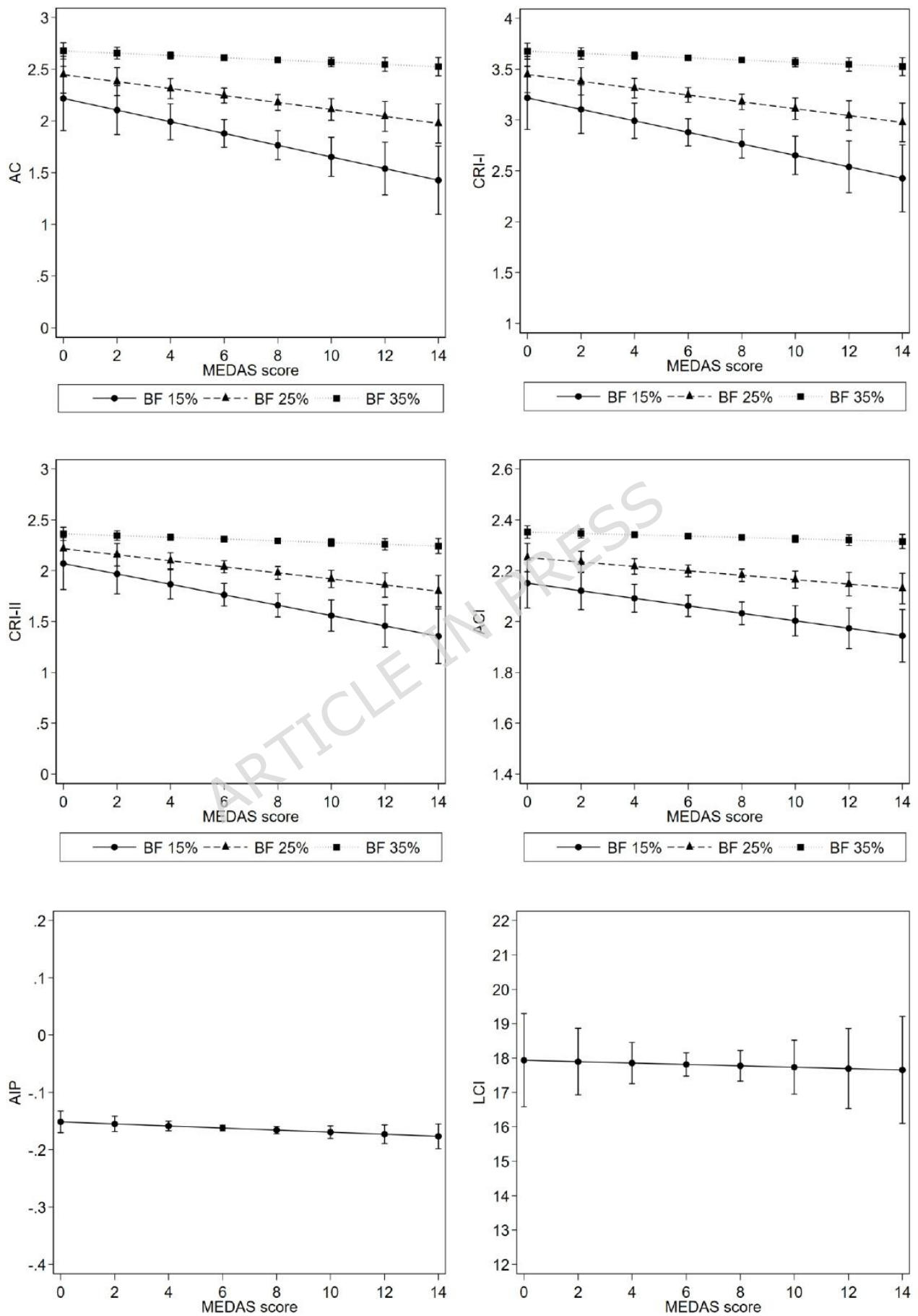


Table 3 presents the main regression models evaluating the association between the MEDAS score and lipid profile parameters. Complete regression results for each individual lipid parameter are provided in Supplementary Tables S9–S12. After adjustment for sex, age, BMI, lifestyle factors, sociodemographic variables, and lipid-lowering therapy (Model 3), a 1-point increase in the MEDAS score was significantly associated with an increase of 0.255 mg/dL in HDL-C cholesterol (95% CI: 0.098, 0.412, $P = 0.001$). This association remained unchanged after further adjustment for body fat percentage. However, a significant interaction between the MEDAS score and body fat percentage was observed for total and LDL-C cholesterol. Specifically, each 1-point increase in the MEDAS score was associated with a reduction of 3.453 mg/dL in total cholesterol (95% CI: –5.911, –0.995, $P = 0.006$) and 3.225 mg/dL in LDL-C cholesterol (95% CI: –5.402, –1.048, $P = 0.004$), but the magnitude of this reduction decreased with increasing body fat percentage. No significant association was found between the MEDAS score and triglyceride levels. Figure 1 displays the effects plots corresponding to Model 4 for HDL-C cholesterol and triglycerides, and to Model 5 for total and LDL-C cholesterol.

Table 3: Association between Mediterranean Diet adherence and lipid profile

	TC M5	HDL-C M4	LDL-C M5	TG M4
Mediterranean score (1-point)	-3.453** [-5.911,-0.995]	0.221** [0.064,0.378]	-3.225** [-5.402,-1.048]	-0.085 [-0.710,0.540]
Body fat (%)	0.649** [0.158,1.140]	-0.285*** [-0.370,-0.200]	0.523* [0.082,0.964]	1.535*** [1.199,1.870]
Mediterranean score # Body fat	0.104** [0.038,0.171]		0.094** [0.035,0.154]	
Sex (Male)	9.905*** [7.394,12.417]	-16.457*** [-17.317,-15.596]	16.362*** [14.096,18.627]	45.705*** [41.789,49.620]
Age (years)	0.780*** [0.704,0.856]	0.143*** [0.115,0.171]	0.646*** [0.578,0.715]	0.189*** [0.080,0.298]
BMI (kg/m ²)	-0.669*** [-0.867,-0.471]	-0.648*** [-0.719,-0.576]	-0.147 [-0.328,0.033]	1.361*** [1.053,1.670]
Physical activity (≥2h/week)	-2.825*** [-4.273,-1.377]	1.276*** [0.737,1.814]	-2.153** [-3.451,-0.854]	-5.882*** [-8.015,-3.748]
Smoking (Ex smoker)	-0.671 [-2.532,1.191]	0.286 [-0.409,0.982]	-2.024* [-3.696,-0.353]	1.917 [-0.816,4.650]
Smoking (Smoker)	0.408 [-1.267,2.084]	-2.341*** [-2.956,-1.726]	1.658* [0.149,3.168]	8.461*** [5.953,10.970]
Education (High school)	1.325 [-1.234,3.885]	0.206 [-0.681,1.093]	1.828 [-0.475,4.132]	-1.401 [-5.297,2.495]
Education (Master degree)	-1.411 [-4.015,1.193]	-0.179 [-1.078,0.719]	0.191 [-2.148,2.530]	-2.783 [-6.714,1.149]
Marital status (Married)	-0.978 [-2.622,0.665]	-0.29 [-0.900,0.320]	-0.521 [-2.001,0.960]	-1.676 [-4.072,0.720]
Marital status (Divorced/ widower)	-0.171 [-2.995,2.654]	0.282 [-0.800,1.363]	-0.84 [-3.366,1.685]	0.527 [-3.594,4.649]
Lipid-lowering therapy (yes)	-18.400*** [-22.721,-14.080]	-0.84 [-2.266,0.585]	-20.387*** [-24.328,-16.445]	9.599** [3.341,15.858]
Constant	166.669*** [149.577,183.761]	86.206*** [83.652,88.760]	83.228*** [68.071,98.385]	-8.334 [-18.066,1.399]

Values are regression coefficients and 95% confidence intervals [in bracket] obtained from multivariate linear regression models.

M4: model adjusted for sex, age, BMI, body fat, smoking status (reference category: never smoker), physical activity (reference category: <2 hours/week), education level (reference category: elementary/middle school), marital status (reference category: single), and use of lipid-lowering therapy (reference category: no)

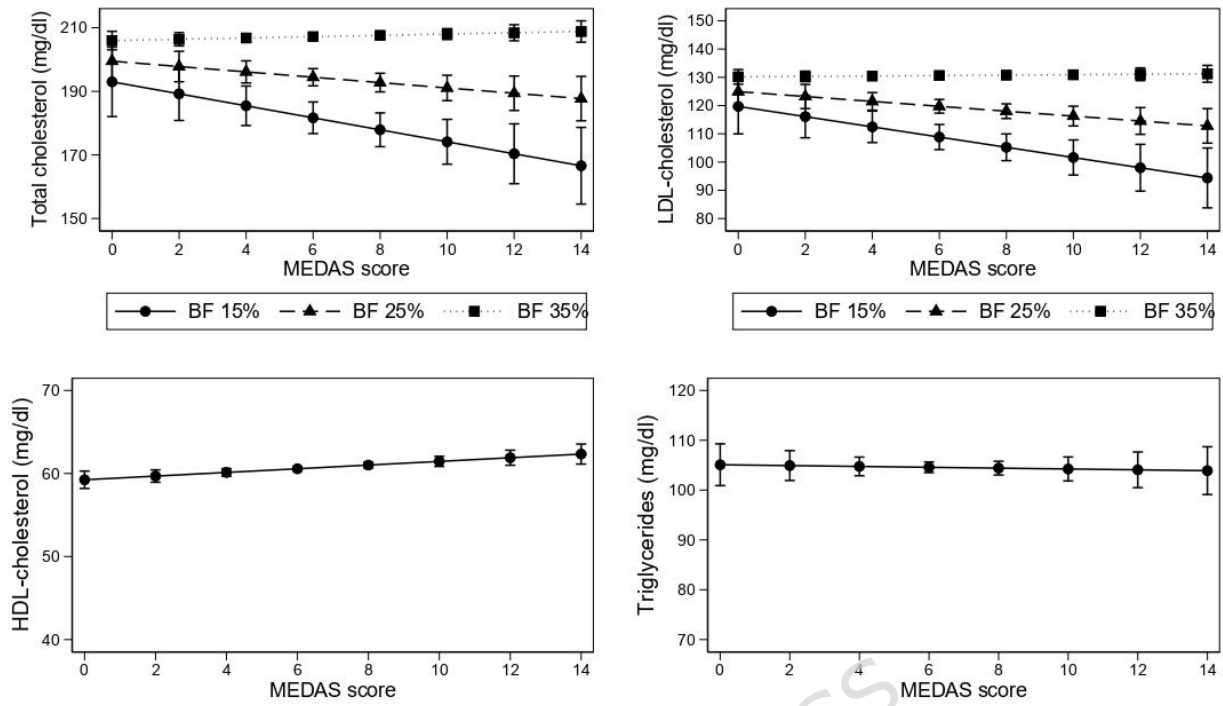
M5: M4 + MEDAS score#body fat interaction

Abbreviations:

TC, total cholesterol; TG, triglycerides.

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Figure 2



Discussion

In the present study, adherence to the Mediterranean diet among individuals seeking a weight-loss program was associated with lower atherogenic indices and a more favorable lipid profile—specifically, lower levels of LDL-C and total cholesterol, and higher HDL-C concentrations. Although the effect sizes observed in the models were numerically small, their cumulative impact across the full range of adherence to the Mediterranean diet is clinically meaningful. When considered over the entire MEDAS scale, these associations translate into substantial reductions in atherogenic indices, supporting the biological relevance of the findings at both the individual and population levels. Interestingly, for several atherogenic indices and lipid parameters, a significant interaction between Mediterranean diet adherence and fat mass was identified, indicating that adiposity may act as an effect modifier. For AIP specifically, the associations disappeared after adjusting for fat mass, suggesting that adiposity may mediate this relationship.

Some previous studies have examined the relationship between diet and atherogenic indices. In a study of 150 women aged 18 to 45 years diagnosed with PCOS, higher adherence to a diet rich in phytochemicals—measured by the Dietary Phytochemical Index (DPI)—was associated with reduced AIP levels [25]. In another study involving 8,870 adults, Ravansar et al. reported a positive association between higher Dietary Inflammatory Index (DII) scores and increased levels of AIP, CRI-I, CRI-II, LCI, and AC [26]. Furthermore, the Isfahan Cohort Study, conducted on 2,256 adults, found that higher scores on the Healthy Lifestyle Score (HLS) were significantly associated with lower levels of AIP, CRI-I, CRI-II, ACI, and AC [27]. To the best of our knowledge, only one study—conducted on a small sample of 51 participants—has reported that the combination of the Mediterranean diet and physical activity was associated with a reduction in AIP [28].

This study, based on a large cohort of participants, further contributes to the literature by demonstrating that adherence to the Mediterranean diet is associated with more favorable atherogenic indices—beyond AIP—and that this effect appears to be mediated by an overall improvement in lipid homeostasis. Specifically, lower levels of total and LDL-C cholesterol, and higher HDL-C concentrations were observed in participants with greater adherence to the Mediterranean diet, consistent with findings from several literature reviews [29].

Although the exact mechanisms remain partially unclear, several synergistic effects among the various components of the Mediterranean diet may explain its beneficial impact on lipid metabolism and atherosclerotic risk. This dietary pattern is rich in unsaturated fatty acids—particularly omega-3 and omega-9—mainly derived from extra virgin olive oil, nuts, seeds, and oily fish. These healthy fats help lower plasma triglycerides and LDL-C cholesterol while increasing HDL-C cholesterol levels [30, 31]. The Mediterranean diet is also high in soluble fiber from fruits, vegetables, legumes, and whole grains, which contributes to reductions in total and LDL-C cholesterol through fermentation by gut microbiota and enhanced bile acid excretion [32, 33]. Moderate red wine consumption adds polyphenols, particularly resveratrol, which may prevent LDL-C oxidation, improve endothelial function, and reduce inflammatory responses involved in atherogenesis [34], while ethanol has been associated with increased HDL-C levels [35]. Importantly, the Mediterranean diet limits foods high in saturated fats—such as red meat, full-fat dairy products, and ultra-processed foods—which are linked to adverse lipid profiles, higher LDL-C/HDL-C ratios, and increased oxidative stress and inflammation [36, 37], as well as excessive caloric intake and fat accumulation that exacerbate cardiovascular risk [38, 39].

Interestingly, for atherogenic indices and lipid parameters, such as AC, CRI-I, CRI-II, ACI, total cholesterol, and LDL-C, a significant interaction between adherence to the

Mediterranean diet and fat mass was observed. Specifically, the strength of the association between adherence to the Mediterranean diet and these markers decreased as fat mass increased. These findings suggest a modifying effect of adipose tissue, whereby excess body fat may progressively diminish the beneficial effects of the Mediterranean diet. This hypothesis is supported by previous studies. In the observational ATTICA study, adherence to the Mediterranean diet was only modestly associated with lower total cholesterol levels among individuals with overweight or obesity [40]. Similar results have been reported in interventional trials, where, in the absence of caloric restriction or weight loss, the Mediterranean diet failed to improve metabolic and lipid profiles [40-42]. Overall, this evidence suggests that the Mediterranean diet should be combined with weight reduction strategies in individuals with overweight or obesity. This finding may be partly explained by the metabolic activity of adipose tissue, especially visceral fat (VAT), which releases free fatty acids, adipokines, and inflammatory cytokines. These substances can impair lipid metabolism by increasing hepatic triglyceride production, reducing HDL-C levels, and promoting LDL-C oxidation [19, 43-46]. As a result, the excess release of these mediators may counteract the beneficial effects of the Mediterranean diet on lipid and atherogenic profiles in individuals with higher body fat. Importantly, excess adiposity is also associated with increased insulin resistance, which represents a key metabolic mechanism linking fat accumulation to adverse lipid alterations. Insulin resistance can further exacerbate hypertriglyceridemia and low HDL-C-cholesterol, thereby potentially counteracting the beneficial effects of the Mediterranean diet on lipid and atherogenic profiles in individuals with higher body fat. It is also plausible that the modifying effect of adiposity reflects underlying sex-related differences in body composition. Women and men differ substantially in total and regional fat distribution, and these differences are known to influence lipid metabolism and atherogenic profiles. In line with this biological framework, the interaction between MEDAS and sex was found to

explain essentially the same variability as the interaction between MEDAS and body fat; moreover, when both interaction terms were included simultaneously in the same model, neither retained statistical significance. This redundancy suggests that sex-related differences in the association between Mediterranean diet adherence and lipid outcomes are largely mediated by differences in body fat, supporting adiposity as the biological driver.

For AIP only, the association with adherence to the Mediterranean diet disappeared after adjusting for fat mass, and no interaction was observed. This suggests that adipose tissue acts as a mediator in the association with the Mediterranean diet. In other words, greater adherence to the Mediterranean diet may lead to reduced fat mass, which in turn contributes to lower AIP values. This hypothesis is supported by a previous study, in which higher adherence to the Mediterranean diet was associated with lower abdominal obesity, particularly with reduced visceral adipose tissue (VAT) [47].

Strengths and limitations

This study has several strengths. The large sample size increased statistical power and the precision of the estimates. In addition, adjustment was made for a wide range of sociodemographic and lifestyle factors that could confound the relationship between diet and lipid outcomes. A further strength of this study is the analysis of lipid parameters and atherogenic indices as continuous variables. This approach captures the full metabolic variability of the cohort and preserves statistical power by avoiding the limitations of categorical stratification. Furthermore, using continuous data ensures that the findings remain robust regardless of potential future updates to clinical cut-offs for dyslipidemia. Nevertheless, future research should explore whether specific dyslipidemic phenotypes exhibit differential responses to the Mediterranean pattern. Finally, a validated questionnaire to screen for adherence to the Mediterranean diet was used.

Nonetheless, this study has several limitations. First, its cross-sectional design prevents any inference of causality. Second, body composition was evaluated through skinfold measurements, which, while not considered the gold standard, offer a practical estimation of fat-free and fat mass. Although more accurate techniques exist, their application in routine clinical settings is limited by high costs, lengthy procedures, and the requirement for highly specialized personnel. In contrast, skinfold assessments are faster, easier to perform, and, in our center, are characterized by very low intra- and inter-operator variability [48]. Third, participants were recruited from a single center and were primarily individuals seeking dietary counseling for weight loss. While this may introduce selection bias and limit generalizability to the general population, the sample reflects a clinically relevant subgroup at increased cardiometabolic risk. Fourth, although the exclusion criteria were implemented to reduce confounding and strengthen internal validity, this approach may limit the generalizability of the findings to individuals with established chronic conditions such as cardiovascular disease or diabetes. Nevertheless, the study population, which includes more than 10,000 individuals actively seeking dietary counseling for weight loss, reflects a relevant and expanding segment of the population at increased cardiometabolic risk. Fifth, the sample was predominantly composed of women, reflecting the higher tendency of females to seek a dietary weight-loss programs [49, 50]. This sex imbalance may limit the generalizability of the findings to a more balanced population. Sixth, dietary adherence was assessed through a self-reported questionnaire, which may be subject to recall and social desirability bias. Moreover, the absence of detailed quantitative dietary data (e.g., total energy intake or macronutrient distribution) prevented exploration of potential confounding or mediation through specific dietary components. Seventh, no formal correction for multiple hypothesis testing was applied, given the exploratory nature of the study and the use of pre-specified regression models. Therefore,

the possibility of type I error cannot be completely excluded. Finally, as with any observational study, potential residual confounding could not be ruled out.

Conclusion

In conclusion, adherence to the Mediterranean diet is associated with more favorable atherogenic and lipid profiles in individuals seeking weight-loss treatment; however, these benefits are significantly modulated by body fat content. Excess adiposity may attenuate or even neutralize the positive effects of the Mediterranean diet on lipid parameters.

Importantly, these findings highlight the need to implement Mediterranean diet–based dietary interventions within integrated clinical strategies that also target body fat reduction. Such an approach may optimize cardiovascular risk management and improve patient care by jointly addressing dietary quality and body composition in individuals at increased cardiometabolic risk.

DECLARATIONS

Ethics approval and consent to participate

The study was conducted in accordance with the guidelines of the Declaration of Helsinki, and each participant read and signed a written informed consent form. The Ethics Committee of the University of Milan approved the study procedures (study protocol No. 23/2016).

Clinical trial number

Not applicable

Consent for publication

Not applicable

Availability of data and materials

The dataset analyzed during the current study is available from the corresponding author on reasonable request.

Competing interest

The authors declare no conflicts of interest

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Author contributions

Conceptualization, A.L.; methodology, A.L., S.B. and A.B.; formal analysis, A.L.; investigation, F.M., S.G., R.D., F.S., G.P., A.F., and A.L.; data curation, A.L.; writing—original draft preparation, F.M. and S.G.; writing—review and editing, A.L. All authors have read and agreed to the published version of the manuscript.

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Figure legend

Figure 1. Association between adherence to the Mediterranean diet and atherogenic indices. The figure presents effect plots illustrating the association between MEDAS score and atherogenic indices, adjusted for sex, age, BMI, smoking status, physical activity, educational level, marital status, use of lipid-lowering therapy, and body fat. For AIP and LCI, the model includes adjustments up to body fat, as the MEDAS # body fat interaction was not significant and therefore excluded. Conversely, for AC, CRI-I, CRI-II, and ACI, the MEDAS # body fat interaction was included in the model due to its statistical significance.

Figure 2. Association between adherence to the Mediterranean diet and lipid parameters. The figure presents effect plots illustrating the association between MEDAS score and lipid parameters, adjusted for sex, age, BMI, smoking status, physical activity, educational level, marital status, use of lipid-lowering therapy, and body fat. For HDL-C and triglycerides, the model includes adjustments up to body fat, as the MEDAS # body fat interaction was not significant and therefore excluded. Conversely, for total cholesterol and LDL-C, the MEDAS # body fat interaction was included in the model due to its statistical significance.